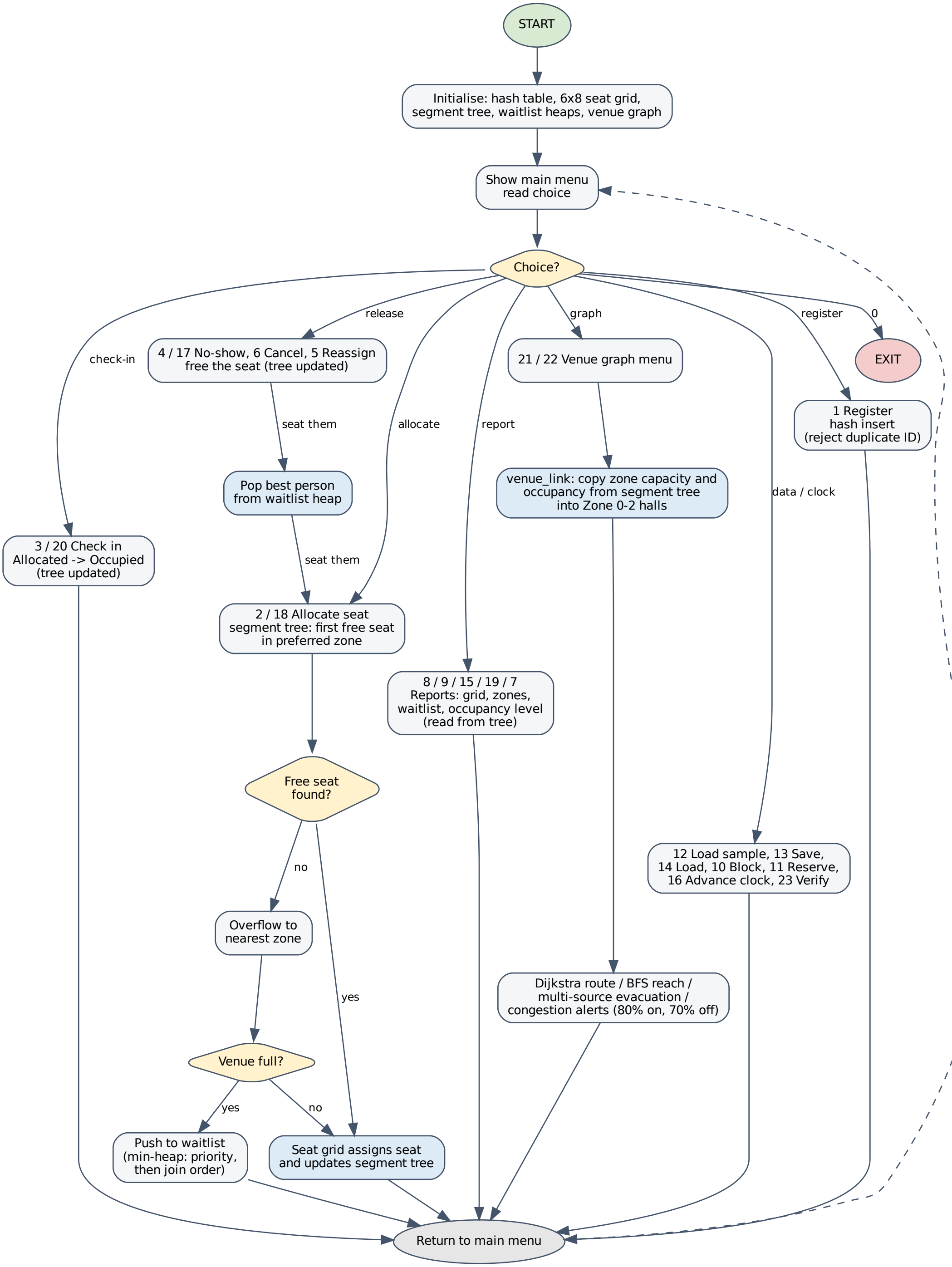
START
Initialise: hash table, 6x8 seat grid, segment tree, waitlist heaps, venue graph
Show main menu read choice
Choice?
check-in
release
graph
register
0
EXIT
4 / 17 No-show, 6 Cancel, 5 Reassign free the seat (tree updated)
21 / 22 Venue graph menu
1 Register hash insert (reject duplicate ID)
seat them
allocate
report
data / clock
3 / 20 Check in Allocated -> Occupied (tree updated)
Pop best person from waitlist heap
venue_link: copy zone capacity and occupancy from segment tree into Zone 0-2 halls
seat them
2 / 18 Allocate seat segment tree: first free seat in preferred zone
8 / 9 / 15 / 19 / 7 Reports: grid, zones, waitlist, occupancy level (read from tree)
Free seat found?
12 Load sample, 13 Save, 14 Load, 10 Block, 11 Reserve, 16 Advance clock, 23 Verify
no
yes
Overflow to nearest zone
Dijkstra route / BFS reach / multi-source evacuation / congestion alerts (80% on, 70% off)
Venue full?
yes
no
Push to waitlist (min-heap: priority, then join order)
Seat grid assigns seat and updates segment tree
Return to main menu